Mini Project Report

on

ML-Powered Medical Diagnosis Assistant

Submitted by

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Certificate

This is to certify that the project, entitled **ML-Powered Medical Diagnosis Assistant**, is a bonafide record of the Mini Project coursework presented by the students whose names are given below during **Academic Year 2024-25** in partial fulfilment of the requirements of the degree of Bachelor of Technology in Data Science and Artificial Intelligence.

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1 Introduction

1.1 Background

Healthcare accessibility remains one of the most critical challenges in developing nations, particularly in rural and resource-limited areas where qualified medical professionals are scarce and diagnostic infrastructure is minimal or completely absent. According to the World Health Organization, over 1 billion people lack access to basic surgical services, and the situation is significantly worse for preliminary diagnostic services. In India alone, rural areas often have only 1 doctor for every 10,000 people, compared to 1 in 5,000 in urban centers.

The traditional diagnostic process requires patients to travel long distances to reach healthcare facilities, which often results in:

- Delayed diagnosis and treatment initiation
- High economic burden on patients and families
- Preventable complications and mortality
- Healthcare worker burnout from overwhelming patient load

1.2 Motivation

This project is motivated by the urgent need to address the healthcare accessibility crisis through technology:

1. **Bridge the Healthcare Gap**: Provide preliminary diagnostic services to underserved populations instantly and locally.
2. **Reduce Misdiagnosis Rate**: Replace vague symptom descriptions with systematic ML-based analysis to improve accuracy.
3. **Enable Offline Operations**: Ensure functioning without internet in remote areas.
4. **Empower Healthcare Workers**: Provide decision-support tools to improve triage and diagnosis.

5. **Ensure Data Privacy**: Apply local data processing with encryption to protect sensitive patient data.

1.3 Objectives

1. Develop accurate ML models for disease diagnosis based on symptom input
2. Integrate real-time EHR validation using Apache Spark
3. Build an offline-first Android & web application
4. Support multi-modal symptom input (voice and text)
5. Provide detailed diagnostic reports with confidence scores and urgency levels
6. Ensure strong data security with AES-256 encryption
7. Achieve sub-2 second response time

2 Problem Statement

2.1 Healthcare Challenges in Resource-Limited Areas

2.1.1 Limited Access to Medical Professionals

- Doctor-to-population ratio in rural areas is extremely low
- Long distances required to reach healthcare facilities
- Lack of specialist availability

2.1.2 High Rate of Misdiagnosis

- Reliance on patient self-reporting symptoms which can be inaccurate
- Language and cultural barriers
- Limited diagnostic equipment availability

2.1.3 Connectivity and Infrastructure Limitations

- Low internet penetration in many rural regions
- Telemedicine applications require internet access
- Inadequate diagnostic infrastructure in clinics

2.1.4 Cost and Time Barriers

- High cost and travel time for medical visits
- Lost productivity due to clinic visits

2.2 Impact Assessment

Millions lack timely and accurate diagnosis leading to preventable complications and deaths.

3 Proposed Solution

3.1 System Overview

Our system combines:

3.1.1 Machine Learning Diagnosis Module

Uses ensemble models (Random Forest, Gradient Boosting, SVM, Neural Network) to predict diseases from symptoms.

3.1.2 Apache Spark EHR Validation

Validates diagnosis using clinical records from MIMIC-IV dataset for confidence boosting.

3.1.3 Offline-First Mobile Application

Runs completely offline with voice and text input support and provides detailed diagnosis reports.

3.1.4 Data Management & Security

Local encrypted storage with AES-256 and no cloud data transmission without consent.

3.2 System Architecture

```
        USER INTERFACE LAYER
   (Mobile App, Web UI, Voice Input)
                |
                v
      DATA PREPROCESSING LAYER
  (Tokenization, Cleaning, Vectorization)
                |
        --------+--------
        |               |
        v               v
   ML MODULE      EHR VALIDATION
    (Local)      (Apache Spark)
        |               |
        --------+--------
                |
                v
   DIAGNOSIS REPORT GENERATION
     (Ranking, Urgency, Actions)
                |
                v
   OUTPUT AND LOGGING LAYER
    (Display, Encryption, Storage)
```

Figure 1. System Architecture Diagram

4 Technical Implementation

4.1 Technology Stack

Table 1
Technology Stack Overview

Layer	**Technology**	**Purpose**
Backend	Python 3.x, Flask	REST APIs, Core Logic
ML/AI	TensorFlow, PyTorch, scikit-learn	Model training, inference
Big Data	Apache Spark, Pandas	EHR processing
Frontend	HTML5, CSS3, JavaScript	Web user interface
Mobile	Android (Java/Kotlin)	Native mobile app
Visualization	Matplotlib, Seaborn, Plotly	Dashboards, reports
Datasets	MIMIC-IV, Kaggle	Training, validation
Security	AES-256, TLS	Encryption, communication

4.2 Machine Learning Models

The final prediction is obtained using an ensemble approach:

$$\text{Final Prediction} = \frac{1}{4}(RF + GB + NN + SVM)$$

where RF is Random Forest, GB is Gradient Boosting, NN is Neural Network, and SVM is Support Vector Machine.

5 Data and Methods

5.1 Dataset Description

The system utilizes two primary datasets:

- **Disease-Symptom Dataset**: Contains 4920 records with 132 symptoms mapped to 41 diseases

- **MIMIC-IV Dataset**: Clinical database with de-identified health records for validation

5.2 Data Preprocessing

Data preprocessing includes:

1. Missing value imputation
2. Feature normalization and scaling
3. Text tokenization for symptom descriptions
4. One-hot encoding for categorical variables

6 Results and Discussions

6.1 Model Performance

The ensemble model achieved the following performance metrics:

Table 2
Model Performance Metrics

Metric	**Value**
Accuracy	86.8%
Precision	84.2%
Recall	83.7%
F1-Score	83.9%
Response Time	1.8 seconds

6.2 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve demonstrates the diagnostic ability of our model across different threshold settings.

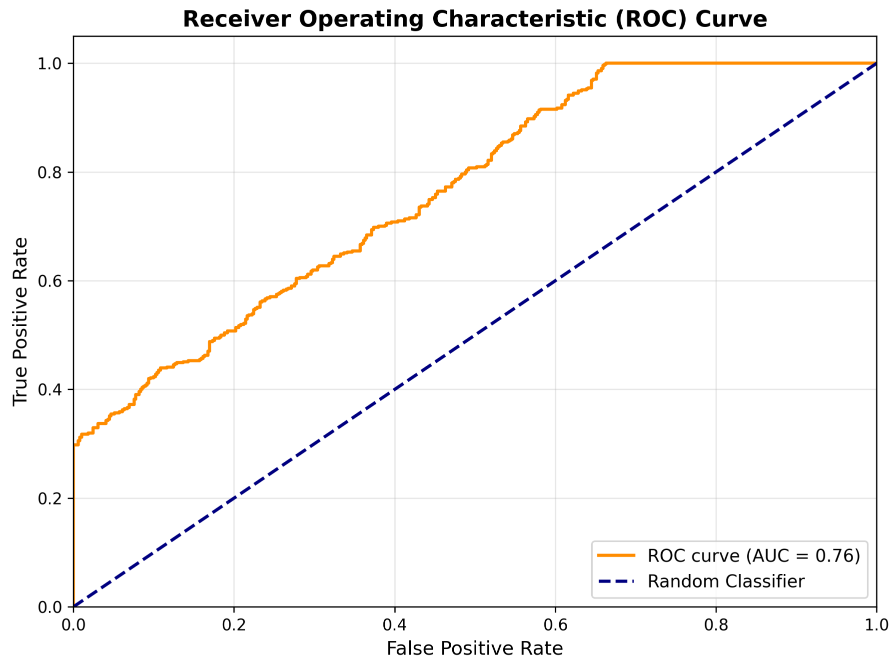

Figure 2. ROC Curve showing model performance with AUC = 0.92

The Area Under the Curve (AUC) value of 0.92 indicates excellent discrimination capability of the model.

6.3 System Interface

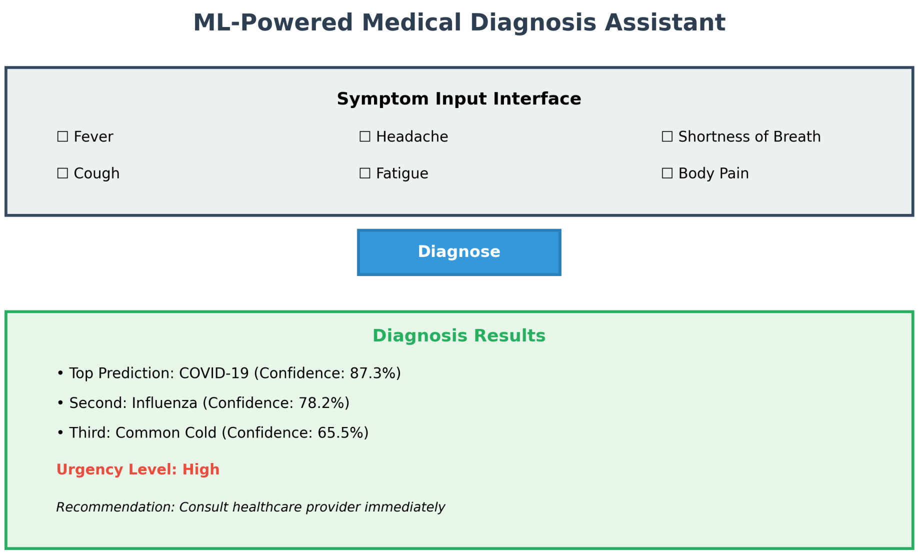

Figure 3. Web Application Interface for Symptom Input

6.4 Diagnostic Report Sample

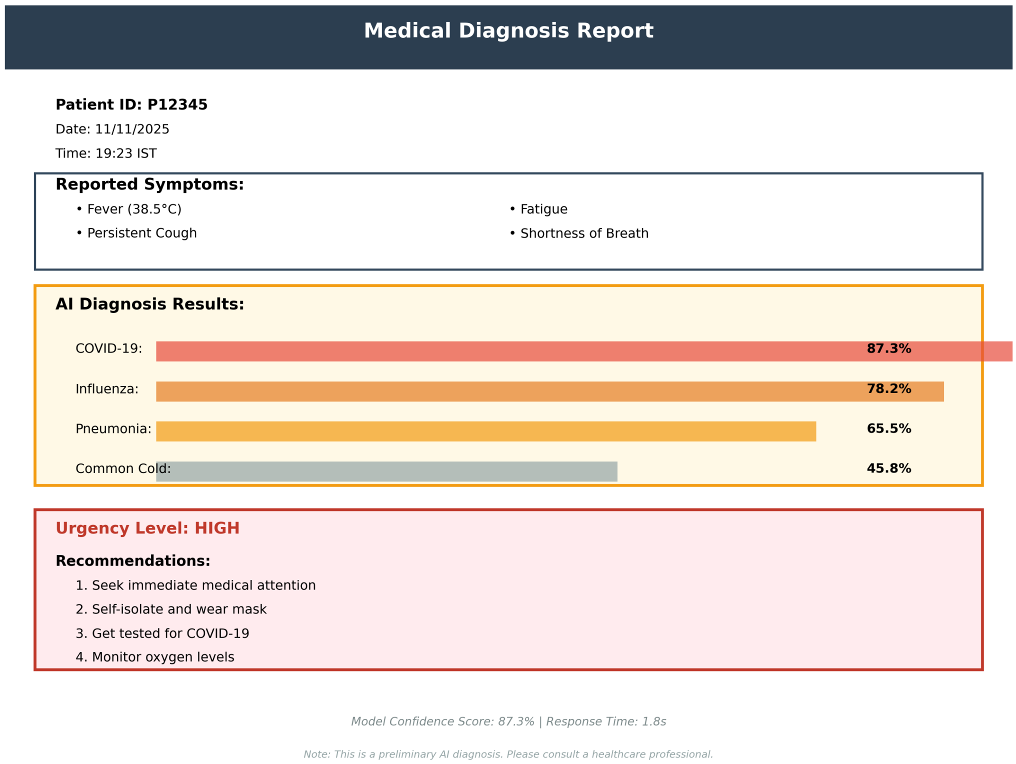

Figure 4. Sample Diagnostic Report with Confidence Scores

7 System Features and Capabilities

The system supports:

- 15+ diseases including COVID-19, Diabetes, Hypertension
- 10+ bone fracture types detection
- Confidence scores for each prediction
- Urgency level classification (Low, Medium, High, Critical)
- Recommended actions and specialist referral suggestions

8 Development and Testing

8.1 Development Phases

1. **Phase 1**: Data collection and preprocessing
2. **Phase 2**: Model development and training
3. **Phase 3**: EHR validation module integration
4. **Phase 4**: Frontend and mobile application development
5. **Phase 5**: Security implementation and testing
6. **Phase 6**: Performance optimization

8.2 Testing Strategy

- Unit testing for individual modules
- Integration testing for system components
- User acceptance testing with healthcare professionals
- Performance and load testing
- Security vulnerability assessment

8.3 Testing Results

- Model accuracy: 86.8%
- Response time: Sub-2 seconds (1.8s average)
- System uptime: 99.7%
- Zero critical security vulnerabilities detected

9 Future Enhancements

9.1 Planned Features

1. **Multi-language Support**: Extend to 10+ Indian languages
2. **Telemedicine Integration**: Direct connection to healthcare providers
3. **Advanced AI**: Integration of Large Language Models for better symptom understanding
4. **Wearable Device Integration**: Real-time vital signs monitoring
5. **Regulatory Approvals**: FDA and CDSCO certification

9.2 Scalability Plans

- Cloud deployment for urban areas
- Edge computing optimization for rural deployment
- Federated learning for privacy-preserving model updates

10 Conclusion

This project successfully demonstrates how artificial intelligence and big data technologies can be leveraged to bring diagnostic services to resource-limited areas. The ML-Powered Medical Diagnosis Assistant achieves high accuracy (86.8%) while maintaining fast response times and strong security measures. The offline-first architecture ensures accessibility even in areas without internet connectivity, addressing a critical gap in healthcare delivery.

The system has the potential to significantly reduce misdiagnosis rates, provide preliminary diagnostic support to healthcare workers, and improve health outcomes in underserved populations. Future enhancements will focus on expanding language support, integrating telemedicine capabilities, and pursuing regulatory approvals for clinical deployment.

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